



Teyzix Core Internship (June Batch)

Task ID: ML-3

Task Title: Smart Electricity Consumption Prediction & Energy Optimization System

Domain: Machine Learning

Difficulty Level: Advanced (Industry-Based)

Assigned Date: 30 June 2026

Submission Deadline: 09 July 2026

Estimated Duration: 10 Days

Project Overview

Efficient energy management has become one of the biggest challenges for households, businesses, and industries due to increasing electricity costs and growing energy demand. Predicting electricity consumption can help users reduce unnecessary energy usage, optimize appliance utilization, and lower monthly electricity bills.

Your task is to develop a **production-ready Smart Electricity Consumption Prediction & Energy Optimization System** using Machine Learning. The system should predict electricity consumption based on user behavior and appliance usage while providing insights into peak usage hours and personalized energy-saving recommendations.

Unlike many machine learning projects, **applicants are required to design and create their own original dataset**. Public datasets from Kaggle, UCI, GitHub, or other online repositories are **strictly prohibited**.

The project should demonstrate the complete machine learning lifecycle, including data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, prediction, and reporting.

Project Objectives

The objective of this task is to evaluate your ability to develop a real-world machine learning solution by implementing:

- Original Dataset Creation
- Data Collection & Validation
- Data Cleaning & Preprocessing
- Feature Engineering
- Exploratory Data Analysis
- Machine Learning Model Development
- Model Evaluation
- Prediction System
- Energy Usage Analysis
- Production-Level ML Workflow

Core Requirements

1. Dataset Creation (Mandatory)

Applicants **must create their own original dataset**.

The dataset must contain at least:

- **500 records**
- **12 or more meaningful features**

Possible features include:

- House Type
- Number of Family Members
- Number of Rooms
- Daily Appliance Usage

- Air Conditioner Usage Hours
- Fan Usage Hours
- Refrigerator Usage
- Washing Machine Usage
- Water Motor Usage
- Lighting Hours
- Outdoor Temperature
- Day of Week
- Season
- Working Days / Holidays
- Daily Electricity Consumption (*Target Variable*)

Applicants may collect data through:

- Household Surveys
- Simulated Energy Records
- Smart Meter Readings (*if available*)
- Realistic Synthetic Data Generation

2. Data Preprocessing

Perform:

- Missing Value Handling
- Duplicate Removal
- Outlier Detection
- Feature Encoding
- Feature Scaling
- Data Normalization

3. Exploratory Data Analysis (EDA)

Perform comprehensive analysis including:

- Statistical Summary
- Feature Distribution
- Correlation Analysis
- Heatmaps
- Histograms
- Box Plots
- Scatter Plots
- Feature Relationships

Provide meaningful business insights from the analysis.

4. Feature Engineering

Implement:

- Feature Selection
- Feature Transformation
- Correlation Analysis
- Feature Importance Evaluation

5. Machine Learning Model Development

Train and compare at least **three different machine learning algorithms**.

Examples include:

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting
- XGBoost
- Support Vector Regression

Compare all models and select the best-performing one.

6. Model Evaluation

Evaluate the model using appropriate regression metrics such as:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Include visual comparisons between actual and predicted values.

7. Prediction System

Develop a simple prediction interface (*CLI, Streamlit, Gradio, or Web Application*) where users can enter electricity usage information and receive:

- Predicted Daily Electricity Consumption
- Estimated Monthly Electricity Usage
- Peak Usage Hours

- Estimated Monthly Electricity Bill
- Energy Efficiency Score

8. Energy Optimization Recommendations

Based on prediction results, the system should generate personalized recommendations such as:

- Best Time to Use High-Power Appliances
- Estimated Monthly Savings
- Peak Consumption Alerts
- Appliance Usage Optimization
- Energy Efficiency Suggestions

9. Report Generation

Generate reports including:

- Dataset Summary
- Model Performance Report
- Energy Consumption Report
- Monthly Usage Forecast
- Recommendation Summary

Reports may be exported as:

- PDF (*Optional*)
- Excel
- CSV

10. Documentation

Document the complete project including:

- Dataset Collection Method
- Feature Description
- Data Preprocessing
- Model Selection
- Performance Comparison
- Challenges Faced
- Future Improvements

Bonus Features (Optional)

Applicants may implement any of the following:

- Weather API Integration
- Electricity Cost Calculator
- Explainable AI (SHAP/LIME)
- Hyperparameter Optimization
- Interactive Dashboard
- Real-Time Prediction
- Smart Home Simulation
- Mobile-Friendly Interface
- Docker Deployment

Technical Requirements

Your project should demonstrate:

- Original Dataset Design
- Data Cleaning
- Feature Engineering
- Exploratory Data Analysis
- Machine Learning Workflow
- Model Comparison
- Regression Analysis
- Modular Code Structure
- Proper Documentation
- Reproducibility

Recommended Technology Stack

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Scikit-learn

- Matplotlib
- Plotly
- XGBoost (*Optional*)

Dashboard (*Optional*)

- Streamlit
- Gradio

Submission Requirements

Applicants must submit:

- Complete Source Code
- Original Dataset (CSV)
- Dataset Documentation
- Jupyter Notebook
- Trained Model
- README.md
- Project Report
- GitHub Repository
- Screenshots
- ZIP File

Evaluation Criteria

Submissions will be evaluated based on the following criteria:

- **Dataset Quality & Originality (20%)** – Quality, relevance, completeness, and originality of the self-created dataset.
- **Data Preprocessing & EDA (20%)** – Effective cleaning, visualization, and meaningful analysis of the dataset.
- **Machine Learning Model (20%)** – Proper model selection, training, comparison, and prediction accuracy.
- **Model Evaluation (15%)** – Appropriate regression metrics, validation techniques, and performance interpretation.
- **Feature Engineering & Business Insights (10%)** – Meaningful feature selection and practical energy consumption insights.
- **Code Quality & Documentation (10%)** – Clean, modular, and well-documented implementation.
- **Innovation & Creativity (5%)** – Additional features, optimization techniques, or enhanced user experience.

Submission Guidelines

- Creating your own dataset is **mandatory**.
- Public datasets from Kaggle, UCI, GitHub, or any other source are **strictly prohibited**.
- Clearly explain how the dataset was collected or generated.
- Ensure the dataset is realistic, balanced, and suitable for machine learning.
- Follow clean coding standards and maintain proper documentation.
- Plagiarism of datasets or machine learning code will result in immediate disqualification.

Expected Learning Outcomes

By completing this task, interns will gain practical experience in:

- Dataset Design & Collection
- Data Cleaning & Feature Engineering
- Exploratory Data Analysis
- Regression-Based Machine Learning
- Model Evaluation
- Energy Consumption Forecasting
- Business Intelligence
- Explainable AI
- Production-Level Machine Learning Workflow

Note

This project addresses a real-world energy management challenge by helping households and businesses predict electricity consumption, estimate future costs, and identify opportunities to reduce energy usage. Applicants are expected to demonstrate originality by creating their own dataset and applying industry-standard machine learning practices. Exceptional submissions may be shortlisted for inclusion in the **TEYZIX CORE** portfolio as showcase projects.
